Thirulok Sundar Mohan Rasu

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Education

University of Michigan — Ann Arbor Aug 2025 – Dec 2026
M.S., Electrical and Computer Engineering GPA: 4.0 / 4.0
Relevant courses: *Matrix Methods for ML & Signal Processing (ECE 551), Probability & Random Processes (ECE 501)*

Indian Institute of Technology (ISM) Dhanbad Dec 2021 – May 2025
B.Tech., Electrical Engineering. Member: *CyberLabs (Machine Learning)* GPA: 8.30 / 10

Skills

Languages: Python, C++, MATLAB, Julia
Signal Processing & ML: Signal Processing, Time-Series Analysis, Frequency-Domain Analysis, Machine Learning, Data Analytics, Feature Engineering, Image Processing
Tools: OpenCV, PyTorch, NumPy, Pandas, SciPy, Matplotlib, Seaborn, HuggingFace, GitHub, Linux

Experience

Carnegie Mellon University (Dr. Arun Balajee Vasudevan) Jan 2024 – Sep 2025
Research Intern

- Engineered end-to-end **data pipelines** for generation, labeling, and evaluation of a **1M+ sample audio dataset**; applied **signal processing techniques** including background noise removal, speech enhancement, and waveform analysis for audio preprocessing
- Fine-tuned **CLAP** audio-language model on the curated dataset achieving **90% accuracy** on synthetic audio detection; paper submitted to **Interspeech 2026** (under review)

Electric Vehicle Center — University of Michigan (Prof. Ziyou Song) Oct 2025 – Present
Graduate Research Assistant

- Built Python data pipelines to process and analyze **electrochemical impedance spectroscopy (EIS)** measurements — **frequency-domain signal characterization** of battery cells; applied physics-informed models with Differential Evolution optimization, achieving **1.7% mean error** across 117 spectra
- Developed **attention-enhanced Seq2Seq** model for **time-series prediction** of battery capacity-fade trajectories, achieving **2–5% MAPE** across 225 cells

HDR Lab — University of Michigan Jan 2026 – Present
Graduate Research Assistant

- Developing **3D depth estimation algorithms** for visuo-tactile sensors; built **iOS app** for automated hardware data collection, enabling large-scale dataset generation for model training and evaluation

Mowito Mar 2023 – Jul 2023
Software Intern

- Built modular **image processing pipeline** using Multi-head Mask R-CNN for object detection and instance segmentation; integrated into production warehouse automation systems with visualization tools
- Implemented **perceptual hashing** algorithm for large-scale image dataset deduplication (**30% size reduction**), improving downstream training efficiency

Selected Projects

Barcode Detection using OpenCV 2024
Built a classic **computer vision pipeline** in Python/OpenCV to detect barcodes on grocery item images; applied **gradient computation, morphological operations, and contour analysis** for barcode localization, then classified the resulting bounding boxes by **barcode readability status** for downstream scanning.

Robot State Estimation: EKF & Particle Filters (GitHub) 2025
Implemented **Extended Kalman Filter** and **Particle Filter** for sensor-based state estimation; benchmarked accuracy and noise-handling trade-offs across five scenarios.

CyScat — Open-Source EM Scattering Simulation Library (GitHub) 2026
Built and shipped an open-source **algorithm library** for electromagnetic scattering simulation; **registered as packages in both Julia and Python registries**; developed using Claude Code for AI-assisted software development.